

SAFE SCAN: The Next Gen Anti-Fraud

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Abstract

Return fraud increasingly includes AI-assisted attacks such as AI-generated "proof" images, as well as physical fraud like duplicate swaps and counterfeits. SAFE SCAN presents a multi-signal verification framework that anchors decisions to a dispatch-time signed reference ("Golden Value") and compares return scans against that reference using embedding similarity, fine-grained micro-features, and fusion scoring. For customer-side uploads, the system gates returns using Gemini-based image authenticity analysis to detect AI-generated or stock-like imagery. An AI Judge aggregates signals into an explainable verdict, risk level, and confidence for operational decisioning.

Keywords: return fraud, product verification, embeddings, fine-grained visual classification, image authenticity, multimodal fusion, explainable decisioning

1. Introduction & Overview

SAFE SCAN blocks return fraud using two complementary ideas: (1) create a **dispatch-time signed reference** (Golden Value) that represents the shipped unit, and (2) run multi-signal verification at return time (embedding + FGVC + FAIM + AI Judge) while also blocking **AI-generated customer uploads** using Gemini.

Primary fraud types

AI-image claim fraud · duplicate swap · counterfeit · item swap

Evidence sources

hash + embeddings + micro-features + fusion score + explainable aggregation

Outputs

verdict (AUTHENTIC / MISMATCH / REVIEW) + risk + confidence + key evidence

Golden Value (signed reference)

Embedding similarity

FGVC micro-features

FAIM fusion

AI Image Detection gate

AI Judge orchestration

2. Method: End-to-End Flow

1) Dispatch / OBD (create truth)

OBD captures multiple angles of the same shipped unit. SAFE-SCAN extracts a stable reference: embedding + micro-features + hash, then cryptographically signs it. This record is the "Golden Value".

```
OBDScanner.jsx
└─ gemini.js:
  extractGoldenValue(images, sku, ...)
    └─ build visual fingerprint
      (embedding + FGVC micro-features)
        └─ imageHash / metadata
          └─ sign + store in Firebase
            (goldenValues)
```

2) Customer return request (gate AI-generated uploads)

Customer-uploaded photos are run through Gemini authenticity detection. If any photo is AI-generated, the return request is blocked/flagged.

```
UserReturnPortal.jsx
└─
  analyzeMultipleImagesWithLogs(images)
    └─ aiImageDetection.js:
      detectAIGeneratedImage(base64)
        └─ Gemini returns:
          isAIGenerated + confidence +
          indicators
```

3) Pickup / FC scan (verify the unit)

The scanned return images are compared against the Golden Value. Multiple independent signals are combined, then the AI Judge makes the final decision with an explanation.

```
Scanner.jsx
└─ gemini.js:
  analyzeReturnScan(returnImages,
    goldenValue, ...)
    └─ embedding similarity
      (cosine/euclidean)
        └─ FGVC micro-features
          └─ optional FAIM fusion
            └─ optional AI image authenticity
              check
                └─ AI Judge: finalVerdict + risk +
                  confidence + evidence
```

Key invariant	Golden Value is verified with signature/hash before any matching is trusted.
Why multi-signal	Fraud must evade several independent checks, not just one threshold.
Explainability	AI Judge provides "why" in plain language + key indicators.

3. Golden Value ("Golden Number"): generation, storage, verification

The "golden number" is a signed record representing the shipped unit's visual identity. It is not a single numeric value — it's a bundle of deterministic signals and model-derived features.

What it contains

```
{
  sku, productName, category,
  imageHash,
  visualEmbedding: [512 numbers],
  fgvcAnalysis: { microFeatures: ... },
  aiMetadata: { labels, text cues, ... },
  capturedAngles: 6,
  createdAt,
  signature: "JWT (HS256)"
}
```

Why signature + hash is essential

The signature prevents tampering: if any stored field is modified later, signature/hash verification fails, and the system rejects the reference as invalid.

Hash purpose	Fast, deterministic integrity fingerprint of the golden payload (SHA-256).
JWT purpose	Cryptographic proof that the Golden Value was generated by SAFE-SCAN and not edited.
Verification gate	Verify signature/hash before using embedding/FGVC for decisioning.

4. AI Image Detection (Gemini): prompt-driven authenticity classification

The goal is to detect whether an uploaded image is a real camera photo vs AI-generated/rendered/stock or heavily manipulated. This is a gated decision in the customer return upload flow.

What Gemini is asked to do

Classification target	REAL_PHOTO vs AI_GENERATED (or equivalent labels).
Evidence requirement	Return indicators used for the decision (human readable).
Confidence	Return numeric confidence so operations can threshold consistently.

Signals evaluated (forensic cues)

Camera pipeline realism sensor noise · JPEG artifacts · DoF blur

Real photos usually show imperfect noise/compression patterns and non-uniform blur.

Physical plausibility shadows · reflections · surface contact

AI often produces inconsistent lighting physics or unnatural contact with surfaces.

Texture micro-variation wear · dust · small scratches · repeated patterns

Diffusion outputs tend to over-smooth or repeat textures at micro scale.

Text/logos typography plausibility · logo edges

AI may render “almost-correct” text/logos with subtle inconsistencies.

Stock-photo signals studio background · perfect lighting

Extremely polished studio shots are suspicious in a customer-return context.

Output shape (used by UI/logging)

```
{
  isAIGenerated: true | false | null,
  confidence: 0..100,
  aiScore: 0..100,
  indicators: [...],
  verdict: "...",
  apiError?: true,
  errorType?: "RATE_LIMITED" | "API_ERROR"
}
```

5. Embedding Similarity: unit-level visual DNA

Embeddings represent images as vectors. Matching becomes a geometry problem: if the returned unit is the same, its embedding should lie close to the Golden Value embedding (after normalization).

What it detects best	item swaps and duplicate swaps where appearance differs at a global level.
Similarity measures	cosine similarity + euclidean distance (combined).
Why normalization	keeps comparisons stable across lighting/scale variation.

6. FGVC Micro-Features: fine-grained counterfeit & duplicate detection

FGVC focuses on fine details: logos, text sharpness, surface integrity, and small differentiators that embeddings sometimes smooth out.

Why it matters	counterfeits often look "close enough" globally but fail micro-feature consistency.
Typical features	logo accuracy · text clarity · edge sharpness · surface defects pattern.
What it outputs	micro-feature scores and a match/mismatch signal.

7. FAIM Fusion: combine vision with operational signals

FAIM (Fused AI for Autonomous Inventory Monitoring) combines vision scores with other signals (e.g., weight) to reduce ambiguity. This is especially useful when images alone are borderline.

Fusion idea	weighted combination of embedding similarity, FGVC, hash similarity, and optional weight checks.
Operational payoff	better separation of "close match" vs "fraud" in challenging conditions.

8. AI Judge: orchestration and final verdict (explainable)

AI Judge is the decision aggregator. It takes multiple algorithm outputs, detects conflicts, assigns risk, and produces an explainable final verdict.

Inputs

```
{
  hashSimilarity,
  visualSimilarity,
  snnSimilarity,
  fgvcAnalysis,
  faimScore,
  aiImageDetection
}
```

Outputs

```
{
  finalVerdict: "AUTHENTIC" | "MISMATCH" | "REVIEW",
  riskLevel: "LOW" | "MEDIUM" | "HIGH",
  confidence: 0..1,
  keyEvidence: [...],
  conflicts: [...]
}
```

This keeps the system panel-ready: you can show not only a verdict, but the chain of evidence and conflict resolution.

9. Decision Policy (thresholds & actions)

SAFE-SCAN uses thresholds to map model signals into business actions (approve, review, block). Exact values are configurable; below is the documented policy.

AI image gate	aiScore 0–30 = real, 30–70 = review, 70–100 = AI-generated → block/flag.
Return verification	use multi-signal agreement; AI Judge resolves conflicts into verdict + risk.
Integrity precondition	Golden Value signature/hash must verify; otherwise reject as untrusted baseline.

10. Reliability / Rate Limits (Gemini): policy and behavior

When Gemini is rate-limited or unavailable, SAFE-SCAN does not silently fall back to weak heuristics. Instead, it returns a structured error state so the UI can ask the operator/user to retry.

Why no fallback	heuristic-only detection is hard to defend and can create unacceptable false decisions.
User guidance	rate limited → "wait and retry"; API error → "service unavailable, retry scan".
Reliability stance	AI authenticity is treated as a required upstream gate, not optional decoration.

11. Limitations & Future Work

Dataset & benchmarking	A full paper version should report results on real return datasets and controlled fraud simulations with clear baselines and ablations.
Robustness	Evaluate robustness to image compression, re-encoding, lighting changes, adversarial perturbations, and "AI-image evasion" attempts.
Privacy & governance	Operational deployments should document data retention, PII handling, and secure key management for signing/verification.
Deployment cost	Report latency and cost per scan, and consider hybrid caching / batching strategies under rate limits.

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Citation

If you reference this work, you can cite it as a technical report / preprint:

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